

MERU, Kenya

WMO: 636950

Lat: 0.083N

Lon: 37.650E

Elev: 5098

StdP: 12.18

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	51.4	53.1	46.3	56.1	61.9	48.1	60.2	60.9	16.9	67.2	14.4	67.1	3.2	250	0.274

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	20.2	80.9	64.4	79.4	64.3	78.2	64.3	68.3	75.0	67.5	74.1	67.0	73.7	5.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.2	117.0	72.0	65.4	113.7	71.2	64.7	110.9	70.6	35.9	75.1	35.3	74.4	34.7	73.8	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.2	10.6	9.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB													
			WB	45.8	71.9	2.3	2.5	44.2	73.7	42.8	75.2	41.6	76.6	39.9	78.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.0	65.3	67.1	68.3	67.8	66.5	64.5	63.2	64.2	66.3	67.7	66.1	65.0
	DBStd	2.50	1.83	1.93	2.11	2.07	1.64	1.82	2.08	2.46	2.20	2.03	1.68	1.66
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	207	17	3	1	2	6	30	62	43	12	3	8	20
	CDD50	5837	474	478	568	533	511	434	411	440	489	550	484	465
	CDD65	570	26	61	104	85	52	15	8	18	51	88	42	20
	CDH74	2087	240	499	601	206	39	7	9	35	165	218	27	41
	CDH80	89	5	29	38	10	0	0	0	2	1	4	0	0
Wind	WSAvg	3.9	3.4	3.6	3.6	3.6	4.4	4.2	4.6	5.0	5.1	4.0	2.7	2.7
Precipitation	PrecAvg	54.0	2.0	1.5	4.6	11.2	4.8	0.4	0.3	0.4	0.7	9.9	12.5	5.7
	PrecMax	116.1	10.4	8.9	20.7	21.8	18.5	1.7	0.9	1.8	4.2	54.6	35.0	17.3
	PrecMin	26.3	0.0	0.0	0.1	3.1	0.4	0.0	0.0	0.0	0.0	0.1	3.8	0.0
	PrecStd	19.6	2.2	2.2	4.7	4.7	3.7	0.4	0.2	0.4	0.9	9.5	6.5	4.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.5	82.8	82.8	81.2	76.7	75.1	75.2	76.9	79.3	80.6	76.3	77.4
		MCWB	62.8	63.8	64.2	66.4	66.1	63.4	62.8	61.8	62.3	64.0	65.0	64.2
	2%	DB	78.3	80.6	80.9	78.4	75.0	73.3	72.9	74.5	77.4	78.3	74.8	75.2
		MCWB	63.6	64.1	64.7	65.7	65.8	63.3	61.5	61.2	62.0	63.6	65.2	64.6
	5%	DB	76.7	78.9	79.2	76.3	73.7	71.9	71.4	72.8	76.0	76.5	73.6	73.8
		MCWB	64.0	64.3	64.9	65.8	65.4	62.9	61.2	60.8	61.8	63.3	65.3	64.7
	10%	DB	75.0	77.3	77.6	74.6	72.5	70.6	69.7	71.0	74.4	74.8	72.3	72.4
		MCWB	64.3	64.4	64.9	65.8	65.1	62.5	60.6	60.4	61.4	63.2	65.1	64.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.4	68.4	71.2	69.0	68.5	66.2	64.6	64.2	65.0	67.2	67.6	68.1
		MCDB	75.3	76.5	77.3	75.8	74.2	71.6	72.3	73.1	74.5	74.4	72.9	73.8
	2%	WB	67.2	67.2	67.7	67.9	67.4	64.8	62.9	62.6	63.8	66.1	66.8	67.1
		MCDB	74.2	76.1	75.8	74.2	72.8	70.9	70.3	71.3	73.9	73.0	72.3	72.4
	5%	WB	66.3	66.3	66.8	67.2	66.5	63.9	62.0	61.8	62.8	65.3	66.2	66.3
		MCDB	73.5	75.6	75.3	73.5	72.0	70.1	69.1	70.1	73.0	72.4	71.7	71.6
	10%	WB	65.4	65.4	65.9	66.5	65.7	63.0	61.2	61.0	62.0	64.6	65.5	65.5
		MCDB	72.7	74.8	74.6	72.8	71.0	68.9	67.7	69.0	71.7	71.8	70.9	70.7
Mean Daily Temperature Range	5% DB	MDBR	21.0	22.6	20.2	15.3	13.7	13.6	13.3	14.6	17.3	15.6	13.9	16.2
		MCDBR	24.3	25.3	23.4	18.0	15.4	15.8	16.6	17.8	19.8	18.3	15.6	18.8
		MCWBR	13.0	12.2	10.3	8.3	7.9	7.6	7.2	7.1	7.4	6.7	8.1	10.5
	5% WB	MCDBR	20.4	21.9	18.9	15.3	13.9	14.2	14.6	16.0	18.1	15.0	14.3	15.8
		MCWBR	12.6	11.9	9.8	8.5	8.0	7.6	7.0	7.0	7.3	6.9	8.2	10.0
Clear-Sky Solar Irradiance	taub	0.349	0.365	0.363	0.338	0.333	0.385	0.394	0.366	0.343	0.340	0.347	0.356	
	taud	2.511	2.451	2.480	2.605	2.616	2.413	2.367	2.449	2.510	2.553	2.551	2.513	
	Ebn at Noon	311	306	304	305	299	279	278	292	305	310	309	307	
	Edh at Noon	35	38	36	31	30	36	38	36	35	34	34	35	
All-Sky Solar Radiation	RadAvg	2012	2123	2067	1888	1858	1681	1617	1726	1997	1943	1760	1835	
	RadStd	131	90	138	94	53	87	107	86	88	73	84	108	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.85	+1.24	N/A	+1.03	+0.70	+0.29	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page